Computer Networks
and Cloud Computing

Data Communication

1.
Which factor primarily affects the **accuracy** of data communication?
A. Bandwidth
B. Noise
C. Latency
D. Throughput

2.
In which mode can communication occur in **both directions simultaneously**?
A. Simplex
B. Half Duplex
C. Full Duplex
D. Parallel

3.
Which of the following is **NOT** a component of data communication?
A. Protocol
B. Medium
C. Compiler
D. Message

4.
Which transmission mode is more **reliable over long distances**?
A. Parallel
B. Serial
C. Full Duplex
D. Half Duplex

5.
Jitter mainly affects which type of application?
A. File transfer
B. Email
C. Real-time video
D. Text processing

6.
Which impairment causes **loss of signal strength**?
A. Noise

B. Distortion
C. Attenuation
D. Delay

7.
Throughput is always:
A. Equal to bandwidth
B. Greater than bandwidth
C. Less than or equal to bandwidth
D. Independent of bandwidth

8.
Which communication mode is used in **walkie-talkies**?
A. Full Duplex
B. Half Duplex
C. Simplex
D. Serial

9.
Which of the following is a **continuous signal**?
A. Digital
B. Binary
C. Analog
D. Packet

10.
Latency refers to:
A. Data loss
B. Delay in transmission
C. Signal strength
D. Transmission speed

11.
Which protocol characteristic defines **structure of data**?
A. Timing
B. Syntax
C. Semantics
D. Flow

12.
Noise in data communication is:
A. Desired signal
B. Amplified signal
C. Unwanted disturbance
D. Encoded data

13.
Parallel transmission is generally used for:
A. Long distances
B. Short distances
C. Wireless systems
D. Satellite communication

14.
Which factor affects **timeliness** the most?
A. Bandwidth
B. Latency
C. Noise
D. Protocol

15.
Distortion occurs when:
A. Signal strength decreases
B. Signal shape changes
C. Signal is delayed
D. Signal is duplicated

16.
Which of the following ensures **rules of communication**?
A. Transmission media
B. Protocol
C. Bandwidth
D. Signal

17.
Which type of communication is used in **TV broadcasting**?
A. Simplex
B. Half Duplex
C. Full Duplex
D. Serial

18.
Bandwidth is best defined as:
A. Actual speed
B. Maximum capacity
C. Delay
D. Signal type

19.
Which signal type is **less prone to noise**?
A. Analog
B. Digital

C. Continuous
D. Wave

20.
Which factor affects **real-time communication quality the most**?
A. Bandwidth
B. Jitter
C. Protocol
D. Message

21.
Which combination best ensures effective communication?
A. High latency + High noise
B. Low jitter + High accuracy
C. Low bandwidth + High delay
D. High attenuation + High distortion

Computer Networks

22.
Which network type is most suitable for a **university campus**?
A. WAN
B. LAN
C. MAN
D. PAN

23.
In which topology does a **single point failure affect the entire network**?
A. Mesh
B. Star
C. Bus
D. Hybrid

24.
Which device operates at the **network layer**?
A. Hub
B. Switch
C. Router
D. Bridge

25.
Which network architecture lacks a **central authority**?
A. Client-Server
B. Hybrid

C. Peer-to-Peer
D. Star

26.
Which transmission medium provides the **highest speed and lowest attenuation**?
A. Twisted Pair
B. Coaxial
C. Fiber Optic
D. Radio Waves

27.
Which topology provides **maximum redundancy**?
A. Bus
B. Star
C. Ring
D. Mesh

28.
Which switching technique is used in the **Internet**?
A. Circuit Switching
B. Packet Switching
C. Message Switching
D. Time Switching

29.
Which device **broadcasts data to all connected devices**?
A. Switch
B. Router
C. Hub
D. Bridge

30.
Which OSI layer is responsible for **end-to-end delivery**?
A. Network
B. Transport
C. Data Link
D. Session

31.
Which network covers a **large geographical area**?
A. LAN
B. MAN
C. WAN
D. PAN

32.

Which topology is **most commonly used in modern LANs**?
A. Bus
B. Ring
C. Star
D. Mesh

33.
Which device connects **different networks using different protocols**?
A. Router
B. Switch
C. Gateway
D. Hub

34.
Which layer in TCP/IP corresponds to **OSI Network layer**?
A. Application
B. Transport
C. Internet
D. Interface

35.
Which topology has **no central node**?
A. Star
B. Bus
C. Ring
D. Hybrid

36.
Which metric measures **actual data transfer rate**?
A. Bandwidth
B. Throughput
C. Latency
D. Delay

37.
Which network type is **city-wide**?
A. LAN
B. WAN
C. MAN
D. PAN

38.
Which device works at the **data link layer**?
A. Router
B. Switch

C. Gateway
D. Repeater

39.
Which switching technique establishes a **dedicated path before transmission**?
A. Packet
B. Circuit
C. Message
D. Hybrid

40.
Which topology is **least scalable**?
A. Mesh
B. Star
C. Bus
D. Hybrid

41.
Which layer is responsible for **logical addressing (IP)**?
A. Data Link
B. Network
C. Transport
D. Physical

Data Link Layer

43.
What is the primary function of the Data Link Layer?
A. End-to-end delivery
B. Node-to-node delivery
C. Routing
D. Encryption
Answer: B

44.
What is the data unit of the Data Link Layer?
A. Packet
B. Segment
C. Frame
D. Bit
Answer: C

45.

Which addressing is used in Data Link Layer?
A. IP Address
B. Logical Address
C. MAC Address
D. Port Address
Answer: C

46.
Which sublayer handles **MAC addressing**?
A. LLC
B. Transport
C. MAC
D. Network
Answer: C

47.
Which error detection technique is most reliable?
A. Parity
B. Checksum
C. CRC
D. Bit stuffing
Answer: C

48.
Flow control ensures:
A. Error-free data
B. Proper speed matching
C. Secure transmission
D. Routing
Answer: B

49.
Stop-and-Wait protocol sends:
A. Multiple frames
B. One frame at a time
C. No frames
D. Infinite frames
Answer: B

50.
Which protocol allows multiple frames before acknowledgment?
A. Stop-and-Wait
B. Sliding Window
C. Parity
D. CRC
Answer: B

51.
Which ARQ retransmits **all frames after error**?
A. Selective Repeat
B. Stop-and-Wait
C. Go-Back-N
D. CRC
Answer: C

52.
Which ARQ retransmits only the **corrupted frame**?
A. Go-Back-N
B. Selective Repeat
C. Stop-and-Wait
D. Parity
Answer: B

53.
What is used to detect errors in frames?
A. MAC
B. CRC
C. IP
D. Port
Answer: B

54.
Which mechanism prevents data overflow at receiver?
A. Error control
B. Flow control
C. Routing
D. Framing
Answer: B

55.
CSMA/CD is used in:
A. Wireless networks
B. Ethernet
C. WAN
D. Satellite
Answer: B

56.
CSMA/CA is used in:
A. Ethernet
B. Wired networks

C. Wireless networks
D. Fiber
Answer: C

57.
Which field marks start/end of frame?
A. Header
B. Trailer
C. Flags
D. Payload
Answer: C

58.
Which sublayer handles error detection and flow control?
A. MAC
B. LLC
C. Network
D. Transport
Answer: B

59.
What happens in **Go-Back-N** after error?
A. Only error frame sent
B. All frames resent
C. No retransmission
D. Connection closed
Answer: B

60.
Which concept allows **shared medium access**?
A. Flow control
B. Access control
C. Error control
D. Framing
Answer: B

61.
What is the role of framing?
A. Routing
B. Error detection
C. Data encapsulation
D. Encryption
Answer: C

62.

Which protocol is least efficient?
A. Sliding Window
B. Selective Repeat
C. Stop-and-Wait
D. Go-Back-N
Answer: C

63.
Which layer ensures reliable link between two nodes?
A. Physical
B. Data Link
C. Network
D. Transport
Answer: B

Network Layer

64.

What is the main responsibility of the Network Layer?
A. Bit transmission
B. Node delivery
C. End-to-end routing
D. Error detection

Answer: C

65.

Which addressing is used in Network Layer?
A. MAC
B. Port
C. IP
D. Frame

Answer: C

66.

Which protocol is used for **routing between networks**?
A. TCP
B. IP
C. HTTP
D. FTP

Answer: B

67.

What does TTL prevent?
A. Packet loss
B. Congestion
C. Infinite looping
D. Delay

Answer: C

68.

Which IP version uses 128 bits?
A. IPv4
B. IPv5
C. IPv6
D. IPv3

Answer: C

69.

Which class of IP is used for **multicast**?
A. A
B. B
C. C
D. D

Answer: D

70.

Which routing algorithm uses **complete network topology**?
A. Distance Vector
B. Link State
C. Flooding
D. Static

Answer: B

71.

Which protocol is used for **internet-level routing**?
A. RIP
B. OSPF
C. BGP
D. ICMP

Answer: C

72.

Fragmentation occurs when:
A. Packet too small
B. Packet too large
C. Network fails
D. IP changes

Answer: B

73.

Which protocol is used for **error reporting**?
A. TCP
B. UDP
C. ICMP
D. ARP

Answer: C

74.

Which routing protocol is **fast and scalable**?
A. RIP
B. OSPF
C. ARP
D. ICMP

Answer: B

75.

Which IP class is used for **large networks**?
A. A
B. B

C. C
D. D

Answer: A

76.

Which device mainly works at Network Layer?
A. Switch
B. Router
C. Hub
D. Bridge

Answer: B

77.

NAT is used to:
A. Encrypt data
B. Translate IP addresses
C. Detect errors
D. Route packets

Answer: B

78.

Which field decreases at each hop?
A. Header
B. TTL
C. Data
D. Address

Answer: B

79.

Which routing uses **hop count**?
A. OSPF
B. BGP
C. RIP
D. ICMP

Answer: C

80.

Which protocol finds **best path dynamically**?
A. Static routing
B. Dynamic routing
C. Manual routing
D. Direct routing

Answer: B

81.

Which part of packet identifies destination?
A. TTL
B. Source IP
C. Destination IP
D. Payload

Answer: C

82.

Which addressing is **global and unique**?
A. MAC
B. IP
C. Port
D. Frame

Answer: B

83.

Which process divides packet into smaller parts?
A. Routing
B. Fragmentation
C. Forwarding
D. Switching

Answer: B

84.

Which protocol is used for **sending ping requests**?
A. TCP
B. UDP
C. ICMP
D. HTTP

Answer: C

Transport Layer

85.

What is the main function of the Transport Layer?
A. Routing
B. Process-to-process delivery
C. Bit transmission
D. Framing

Answer: B

86.

Which protocol is **connection-oriented**?
A. UDP
B. TCP
C. IP
D. HTTP

Answer: B

87.

Which protocol is **faster but unreliable**?
A. TCP
B. UDP
C. IP
D. ARP

Answer: B

88.

Which unit is used in Transport Layer?
A. Frame

B. Packet
C. Segment
D. Bit

Answer: C

89.

Which mechanism ensures correct application receives data?
A. Routing
B. Port addressing
C. Framing
D. Switching

Answer: B

90.

Which protocol is used in **video streaming**?
A. TCP
B. UDP
C. HTTP
D. FTP

Answer: B

91.

What does TCP guarantee?
A. Speed
B. Reliability
C. Low delay
D. Broadcast

Answer: B

92.

Which process divides data into segments?
A. Routing
B. Segmentation
C. Switching
D. Framing

Answer: B

93.

Which concept controls network overload?
A. Flow control
B. Congestion control
C. Error control
D. Routing

Answer: B

94.

Which field identifies application process?
A. IP
B. MAC
C. Port
D. TTL

Answer: C

95.

Which protocol uses **three-way handshake**?
A. UDP
B. TCP
C. IP
D. ICMP

Answer: B

96.

What is the first step in TCP handshake?
A. ACK
B. SYN
C. FIN
D. RST

Answer: B

97.

Which protocol does NOT ensure order?
A. TCP
B. UDP
C. HTTP
D. FTP

Answer: B

98.

Which port range is for well-known services?
A. 1000–2000
B. 0–1023
C. 2000–4000
D. 4000–8000

Answer: B

99.

Which mechanism prevents receiver overflow?
A. Routing
B. Flow control
C. Error control
D. Fragmentation

Answer: B

100.

Which protocol is used for **email transfer reliability**?
A. UDP
B. TCP
C. IP
D. ARP

Answer: B

101.

Which technique allows multiple applications to share connection?
A. Routing
B. Multiplexing

C. Switching
D. Encoding

Answer: B

102.

Which process delivers data to correct application?
A. Multiplexing
B. Demultiplexing
C. Routing
D. Fragmentation

Answer: B

103.

Which protocol is best for **online gaming**?
A. TCP
B. UDP
C. FTP
D. SMTP

Answer: B

104.

Which mechanism ensures lost packets are resent?
A. Flow control
B. Error control
C. Routing
D. Switching

Answer: B

105.

Which TCP feature adjusts sending rate dynamically?
A. Routing
B. Congestion control
C. Fragmentation
D. Encoding

Answer: B

Application Layer

106. Which layer interacts with user?
   A. Transport
   B. Network
   C. Application
   D. Data Link
   Answer: C

107. HTTP is used for:
   A. Email
   B. Web browsing
   C. File transfer
   D. Routing
   Answer: B

108. Which protocol translates domain to IP?
   A. FTP
   B. DNS
   C. HTTP
   D. SMTP
   Answer: B

109. SMTP is used for:
   A. Receiving email
   B. Sending email
   C. Routing
   D. Encryption
   Answer: B

110. FTP is used for:
   A. Browsing
   B. File transfer
   C. Email
   D. Routing
   Answer: B

111. Which protocol is secure HTTP?
   A. FTP
   B. HTTPS
   C. SMTP
   D. DNS
   Answer: B

112. POP3 is used for:
A. Sending mail
B. Receiving mail
C. Routing
D. Switching
Answer: B

113. HTTP is:
A. Stateful
B. Stateless
C. Secure
D. Connectionless always
Answer: B

114. Which protocol resolves IP to domain?
A. DNS
B. Reverse DNS
C. FTP
D. HTTP
Answer: B

115. IMAP differs from POP3 because it:
A. Deletes mail
B. Stores on server
C. Sends mail
D. Encrypts data
Answer: B

116. URL stands for:
A. Uniform Resource Locator
B. Universal Resource Link
C. Unified Routing Layer
D. User Resource Locator
Answer: A

117. Which protocol uses port 80?
A. FTP
B. HTTP
C. SMTP
D. DNS
Answer: B

118. Which protocol uses port 443?
A. HTTP
B. HTTPS
C. FTP

D. SMTP
Answer: B

119. DNS works on which model?
A. P2P
B. Client-Server
C. Hybrid
D. Mesh
Answer: B

120. Which protocol is fastest for file transfer?
A. FTP
B. HTTP
C. SMTP
D. DNS
Answer: A

121. Which protocol is used in web APIs?
A. FTP
B. HTTP
C. SMTP
D. ARP
Answer: B

122. Which layer handles web services?
A. Transport
B. Application
C. Network
D. Physical
Answer: B

123. Which protocol is used for name resolution?
A. DNS
B. FTP
C. SMTP
D. HTTP
Answer: A

124. Which protocol is used for downloading emails?
A. SMTP
B. POP3
C. HTTP
D. TCP
Answer: B

125. Which protocol maintains server copy of emails?
A. POP3

B. IMAP
C. SMTP
D. FTP
Answer: B

126. Which protocol is stateless?
A. FTP
B. HTTP
C. SMTP
D. DNS
Answer: B

7 - Wireless Networks

 Wireless uses:
A. Fiber
B. Radio waves
C. Copper
D. Coaxial
Answer: B

 Which is wireless tech?
A. Ethernet
B. Wi-Fi
C. Fiber
D. DSL
Answer: B

 Bluetooth is used for:
A. Long distance
B. Short range
C. Satellite
D. WAN
Answer: B

 Which network is cellular?
A. LAN
B. 5G
C. Bus
D. Ring
Answer: B

 Main issue in wireless:
A. Speed
B. Interference
C. Cost
D. Routing
Answer: B

 Wi-Fi uses which medium?
A. Wire
B. Radio
C. Fiber
D. Coaxial
Answer: B

 Which is most secure?
A. Open Wi-Fi
B. WPA2
C. Bluetooth
D. Infrared
Answer: B

 Wireless bandwidth is:
A. Unlimited
B. Limited
C. Fixed
D. Infinite
Answer: B

 Which tech uses infrared?
A. Remote
B. Wi-Fi
C. LAN
D. Fiber
Answer: A

 Cellular networks use:
A. Towers
B. Switches
C. Routers
D. Bridges
Answer: A

 Which is PAN?
A. Wi-Fi
B. Bluetooth
C. WAN

D. MAN
Answer: B

 Wireless suffers from:
A. Noise
B. No delay
C. No errors
D. Infinite speed
Answer: A

 Which is fastest wireless?
A. 2G
B. 3G
C. 5G
D. Bluetooth
Answer: C

 Which uses CSMA/CA?
A. Ethernet
B. Wi-Fi
C. Fiber
D. DSL
Answer: B

 Wireless security uses:
A. Encryption
B. Routing
C. Framing
D. Switching
Answer: A

 Which has highest range?
A. Bluetooth
B. Wi-Fi
C. Cellular
D. Infrared
Answer: C

 Which is low power?
A. Wi-Fi
B. Bluetooth
C. 5G
D. Fiber
Answer: B

 Which suffers attenuation?
A. Wired

B. Wireless
C. Both
D. None
Answer: C

 Which tech used in IoT?
A. Bluetooth
B. Wi-Fi
C. Both
D. None
Answer: C

 Wireless communication is:
A. Guided
B. Unguided
C. Wired
D. Optical
Answer: B

 Which uses frequency spectrum?
A. Wired
B. Wireless
C. LAN
D. Bus
Answer: B

8 - Cloud Computing

 Cloud provides:
A. Hardware
B. Services over internet
C. LAN
D. Routing
Answer: B

 SaaS provides:
A. Software
B. Hardware
C. Network
D. Storage only
Answer: A

 IaaS provides:
A. Software
B. Infrastructure

C. Platform
D. Apps
Answer: B

 PaaS provides:
A. Apps
B. Platform
C. Network
D. Router
Answer: B

 Public cloud is:
A. Private
B. Shared
C. Offline
D. Local
Answer: B

 Hybrid cloud is:
A. Single
B. Mixed
C. Offline
D. Wired
Answer: B

 Example of SaaS:
A. Gmail
B. Router
C. Switch
D. LAN
Answer: A

 Cloud uses:
A. Internet
B. LAN
C. Bus
D. Ring
Answer: A

 Which is scalable?
A. Cloud
B. LAN
C. Bus
D. Ring
Answer: A

 Cloud reduces:
A. Cost
B. Speed
C. Data
D. Protocol
Answer: A

 Which is IaaS example?
A. AWS EC2
B. Gmail
C. YouTube
D. DNS
Answer: A

 Which is PaaS example?
A. Heroku
B. Gmail
C. Router
D. FTP
Answer: A

 Which is private cloud?
A. Shared
B. Organization-specific
C. Public
D. Internet
Answer: B

 Cloud storage is:
A. Local
B. Remote
C. Wired
D. Offline
Answer: B

 Which provides apps directly?
A. IaaS
B. PaaS
C. SaaS
D. LAN
Answer: C

 Cloud depends on:
A. Internet
B. Router
C. Switch

D. Bus
Answer: A

 Which is benefit?
A. Less scalability
B. High cost
C. Flexibility
D. Slow
Answer: C

 Which is not cloud?
A. AWS
B. Azure
C. LAN
D. Google Cloud
Answer: C

 Which is disadvantage?
A. Cost
B. Security risk
C. Speed
D. Access
Answer: B

 Cloud enables:
A. Remote access
B. Local only
C. No sharing
D. Isolation
Answer: A

 Which model gives full control?
A. SaaS
B. PaaS
C. IaaS
D. DNS
Answer: C

- Network Security (Networks Perspective)

 CIA stands for:
A. Control Integrity Access
B. Confidentiality Integrity Availability
C. Central Internet Access

D. Control Internet Access
Answer: B

 DoS attack affects:
A. Confidentiality
B. Availability
C. Integrity
D. Authentication
Answer: B

 Phishing targets:
A. Network
B. Users
C. Hardware
D. Routing
Answer: B

 Encryption ensures:
A. Speed
B. Security
C. Delay
D. Routing
Answer: B

 Firewall is used for:
A. Routing
B. Filtering traffic
C. Switching
D. Framing
Answer: B

 Antivirus protects from:
A. Routing
B. Malware
C. Switching
D. Framing
Answer: B

 Integrity ensures:
A. Data unchanged
B. Data secret
C. Data available
D. Data fast
Answer: A

 Confidentiality ensures:
A. Access control

B. Data secret
C. Speed
D. Routing
Answer: B

 Availability ensures:
A. Access anytime
B. Encryption
C. Routing
D. Framing
Answer: A

 MITM attack means:
A. Middle intercept
B. Routing error
C. Switching
D. Framing
Answer: A

 VPN provides:
A. Speed
B. Secure tunnel
C. Routing
D. Switching
Answer: B

 Authentication verifies:
A. Data
B. Identity
C. Speed
D. Path
Answer: B

 Authorization means:
A. Identity
B. Permission
C. Routing
D. Switching
Answer: B

 Hashing ensures:
A. Integrity
B. Confidentiality
C. Speed
D. Routing
Answer: A

 SSL/TLS provides:
A. Encryption
B. Routing
C. Switching
D. Framing
Answer: A

 Which attack floods network?
A. Phishing
B. DoS
C. MITM
D. Spoofing
Answer: B

 Spoofing means:
A. Fake identity
B. Routing
C. Delay
D. Packet loss
Answer: A

 Which protects wireless?
A. WPA
B. FTP
C. HTTP
D. DNS
Answer: A

 Malware includes:
A. Virus
B. Worm
C. Trojan
D. All
Answer: D

 IDS detects:
A. Intrusions
B. Routing
C. Switching
D. Framing
Answer: A

 Which ensures secure communication?
A. Encryption
B. Routing
C. Switching

D. Framing
Answer: A

10 - Next Generation Networks

190. NGN stands for:
   A. New Global Network
   B. Next Generation Network
   C. Network Global Node
   D. New General Network
   Answer: B

191. NGN is based on:
   A. Circuit
   B. IP
   C. Analog
   D. Manual
   Answer: B

192. 5G provides:
   A. Low speed
   B. High latency
   C. High speed
   D. No connectivity
   Answer: C

193. IoT connects:
   A. Devices
   B. Routers
   C. Switches
   D. Servers
   Answer: A

194. SDN separates:
   A. Hardware
   B. Control & data plane
   C. Routing
   D. Switching
   Answer: B

195. NFV virtualizes:
   A. Hardware
   B. Network functions
   C. Routing

D. Switching
Answer: B

196. NGN supports:
A. Voice
B. Data
C. Video
D. All
Answer: D

197. Which is future tech?
A. 2G
B. 3G
C. 5G
D. LAN
Answer: C

198. IoT uses:
A. Sensors
B. Routers
C. Switches
D. Cables
Answer: A

199. NGN improves:
A. Speed
B. Efficiency
C. Integration
D. All
Answer: D

200. Which is programmable network?
A. SDN
B. LAN
C. WAN
D. MAN
Answer: A

201. NFV reduces:
A. Cost
B. Speed
C. Data
D. Delay
Answer: A

202. NGN replaces:
A. Traditional networks

B. LAN
C. WAN
D. Router
Answer: A

203. Which supports real-time apps?
A. NGN
B. LAN
C. Bus
D. Ring
Answer: A

204. IoT depends on:
A. Internet
B. LAN
C. Bus
D. Ring
Answer: A

205. Which tech enables automation?
A. IoT
B. LAN
C. Bus
D. Ring
Answer: A

206. 5G latency is:
A. High
B. Low
C. Medium
D. None
Answer: B

207. SDN improves:
A. Flexibility
B. Cost
C. Control
D. All
Answer: D

208. NGN integrates:
A. Voice
B. Data
C. Multimedia
D. All
Answer: D

209. IoT devices are:

A. Smart

B. Dumb

C. Offline

D. Wired only

Answer: A

210. NGN uses:

A. Packet switching

B. Circuit switching

C. Manual switching

D. Analog

Answer: A